

Systems Integration: Core Concepts and Fundamentals

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Cue / Keyw ords	Notes
System	Process of bringing different systems together into one unified, interoperable system that facilitates communication across the network. Poor integration is called "Spaghetti Integration".
Integration	
Enterprise	Strategic approach achieving interoperability between IT applications (internal/external). Business-driven, involves multiple stakeholders, strategic view rather than localized.
Integration	
Business	- Information Aggregation : Single view of multiple sources. - Single point of entry : Reduced manual/duplicate data entry. - Process inefficiency : Reduced effort/time. - Web channel & Supplier integration : External access and optimized supply chain.
Problems	
Solved	
Integration	- Horizontal Integration (ESB) : Subsystem acts as interface between all others. - Vertical Integration : Creation of silos. - Star Integration : Point-to-point connections. - Common data format : Application-independent format/transformation service.
Methods	
Integration	- EAI (Enterprise Application Integration) : Internal workflows. - DI (Data Integration) : Consolidating raw data for analytics. - EDI (Electronic Data Interchange) : Secure standardized document exchange between external partners.
Types	
Implementation	1. Requirements Gathering 2. Data Analysis 3. Design Architecture 4. Design Systems integration 5. System Implementation
Steps	
Assessment	Necessary when there are many isolated systems, duplicate data entries, inconsistent data, difficult information extraction, batch uploads, or need for real-time information collation.
Needs	

Summary

Systems integration combines isolated sub-systems into a unified, interoperable whole, primarily to resolve business inefficiencies like data duplication and disjointed workflows. Unlike basic integration, Enterprise Integration is a strategic, business-driven approach. Implementation typically involves standardized phases, relying on methods like Horizontal (ESB), Vertical, or Star integration, and categories like EAI, DI, and EDI depending on the specific integration scope.